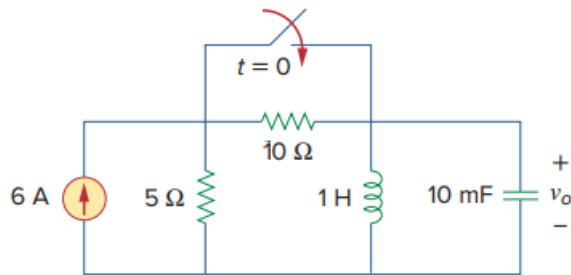


Problem

Find the output voltage $v_o(t)$ in the circuit of Fig. 8.94.

Figure 8.94:



Step-by-step solution

Step 1 of 9

Refer to circuit diagram in Figure 8.94 in the text book.

Initially switch in the circuit is open for long time and circuit reaches steady state. Under steady state capacitor acts as open circuit and inductor acts as short circuit. The circuit diagram for $t < 0$ is shown in Figure 1.

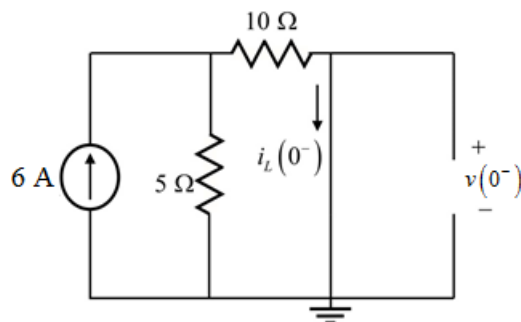


Figure 1

Step 2 of 9

The value of the capacitor voltage $v(0^-)$ is 0 V since both the terminals are shorted.

$$v(0^-) = 0 \text{ V}$$

Calculate the value of the current $i(0^-)$ using current division rule.

$$\begin{aligned} i(0^-) &= (6) \left(\frac{5}{5+10} \right) \\ &= (6) \left(\frac{5}{15} \right) \\ &= 2 \text{ A} \end{aligned}$$

Therefore, the value of the current $i(0^-)$ is 2 A.

Step 3 of 9

For $t > 0$ switch in the circuit is closed, the modified circuit is shown in Figure 2.

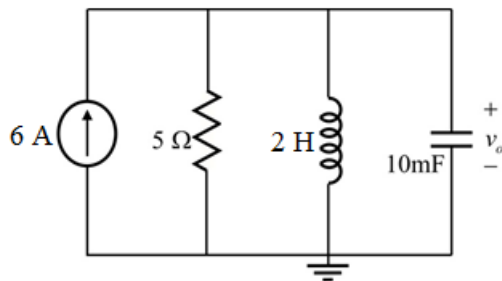


Figure 2

Step 4 of 9

Calculate the neper frequency.

$$\alpha = \frac{1}{2RC}$$

Substitute 5Ω for R , 10 mF for C .

$$\begin{aligned}\alpha &= \frac{1}{2RC} \\ &= \frac{1}{2(5)(10 \times 10^{-3})} \\ &= 10\end{aligned}$$

Determine the value of ω_0 .

$$\omega_0 = \frac{1}{\sqrt{LC}}$$

Substitute 2 H for L , 10 mF for C .

$$\begin{aligned}\omega_0 &= \frac{1}{\sqrt{LC}} \\ &= \frac{1}{\sqrt{(2)(10 \times 10^{-3})}} \\ &= \frac{1}{0.1414} \\ &= 7.07 \text{ rad/s}\end{aligned}$$

Step 5 of 9

The system is in overdamped condition because $\alpha > \omega_0$.

The voltage across the capacitor expression is,

$$v_o(t) = V_s + A_1 e^{s_1 t} + A_2 e^{s_2 t}$$

At $t = \infty$ capacitor acts as short circuit and inductor acts as open circuit. The final voltage across the capacitor is,

$$V_s = 0 \text{ V}$$

Calculate the roots of the characteristic equation.

$$\begin{aligned}s_{1,2} &= -\alpha \pm \sqrt{\alpha^2 - \omega_0^2} \\ &= -10 \pm \sqrt{10^2 - 7^2} \\ &= -10 \pm 7.1414 \\ &= -2.8586, -17.1414\end{aligned}$$

Step 6 of 9

The voltage across the capacitor expression is,

$$v_o(t) = V_s + A_1 e^{s_1 t} + A_2 e^{s_2 t}$$

Substitute 0 V for V_s , -2.8586 for s_1 and -17.1414 for s_2 .

$$v_o(t) = A_1 e^{-2.8586t} + A_2 e^{-17.1414t} \dots\dots (1)$$

Substitute 0 for t .

$$v_o(0) = A_1 e^{-2.8586(0)} + A_2 e^{-17.1414(0)}$$

$$0 = A_1 + A_2 \quad \left(\text{since } v(0^-) = 2 \text{ A} \right)$$

$$A_1 + A_2 = 0 \dots\dots (2)$$

Step 7 of 9

Draw the circuit for $t = 0^+$ is shown in Figure 3.

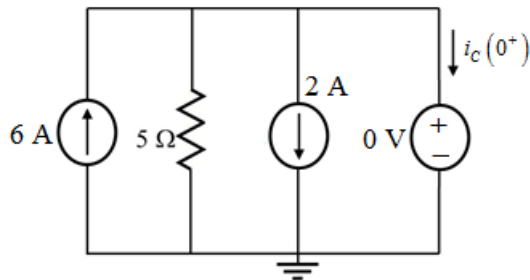


Figure 3

Step 8 of 9

From the circuit shown in Figure 3, the voltage across the inductor is,

$$-6 + \frac{0}{5} + 2 + i_c(0^+) = 0$$

$$i_c(0^+) = 6 - 2$$

$$(10 \times 10^{-3}) \frac{dv_o(0^+)}{dt} = 4$$

$$\frac{dv_o(0^+)}{dt} = 400 \text{ V/s}$$

Recall equation (1).

$$v_o(t) = A_1 e^{-2.8586t} + A_2 e^{-17.1414t}$$

Differentiate the equation with respect to t .

$$\frac{dv_o(t)}{dt} = -2.8586 A_1 e^{-2.8586t} - 17.1414 A_2 e^{-17.1414t}$$

Substitute 0 for t .

$$\frac{dv_o(0)}{dt} = -2.8586 A_1 e^{-2.8586(0)} - 17.1414 A_2 e^{-17.1414(0)}$$

$$400 = -2.8586 A_1 - 17.1414 A_2 \dots\dots (3)$$

Step 9 of 9

Solve equations (2) and (3).

$$A_1 = 28$$

$$A_2 = -28$$

Recall equation (1).

$$v_o(t) = A_1 e^{-2.8586t} + A_2 e^{-17.1414t}$$

Substitute 28 for A_1 and -28 for A_2 .

$$v_o(t) = 28e^{-2.8586t} - 28e^{-17.1414t} \text{ V}$$

Therefore, the voltage across the capacitor for $t > 0$ is $\boxed{28e^{-2.8586t} - 28e^{-17.1414t} \text{ V}}$.